

AI Assistant Coding Assignment-17.4

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Lab 17: AI for Data Processing: Data Cleaning and Preprocessing Scripts

Task 1 – Smart City Traffic Sensor Data Cleaning

Scenario

A Smart City project collects real-time traffic sensor data from multiple intersections. However, the dataset contains inconsistent formats, missing values, and corrupted readings.

Task

Use AI to generate a Python script to clean and preprocess the traffic dataset.

Instructions

- Handle missing values in:
 - o vehicle_count
 - o avg_speed
- Remove duplicate sensor readings
- Convert timestamp column into proper datetime format
- Remove invalid readings (e.g., negative speed or vehicle count)
- Standardize intersection names (uppercase format)

Expected Output

- Cleaned traffic dataset
- Valid sensor readings only
- Proper datetime column
- No duplicates or invalid values

Prompt:

Dataset contains traffic sensor readings with inconsistencies like missing values, duplicates, invalid readings, and unstandardized intersection names.

AI Generated Code:

```
import pandas as pd
import numpy as np

# Load dataset
traffic_df = pd.read_csv("/content/traffic.csv")

# Convert timestamp to datetime
traffic_df['DateTime'] = pd.to_datetime(traffic_df['DateTime'], errors='coerce')

# Remove duplicate sensor readings
traffic_df = traffic_df.drop_duplicates()
```

```

missing values traffic_df['Vehicles'] =
traffic_df['Vehicles'].fillna(traffic_df['Vehicles
'].median())
# The 'avg_speed' column does not appear in the current dataframe, commenting this line out.
# traffic_df['avg_speed'] = traffic_df['avg_speed'].fillna(traffic_df['avg_speed'].median())

# Remove invalid readings traffic_df = traffic_df[(traffic_df['Vehicles'] >= 0)] # Removed
condition for 'avg_speed' as it is not present

# Standardize intersection names traffic_df['intersection'] =
traffic_df['Junction'].astype(str).str.upper() # Assuming 'Junction' is the intersection column

# Save cleaned dataset traffic_df.to_csv("traffic_data_cleaned.csv",
index=False) print("Cleaned Traffic Dataset Ready")

```

The screenshot shows a Jupyter Notebook environment. The left pane contains a Python script for cleaning a traffic dataset. The script imports pandas and numpy, loads a CSV file, converts timestamps, removes duplicates, handles missing values, removes invalid readings, standardizes intersection names, and saves the cleaned dataset. The right pane shows a Gemini AI chat window with a response confirming the data cleaning process and providing suggestions for further analysis.

```

[9] ✓ On
import pandas as pd
import numpy as np

# Load dataset
traffic_df = pd.read_csv("/content/traffic.csv")

# Convert timestamp to datetime
traffic_df['DateTime'] = pd.to_datetime(traffic_df['DateTime'], errors='coerce')

# Remove duplicate sensor readings
traffic_df = traffic_df.drop_duplicates()

# Handle missing values
traffic_df['Vehicles'] = traffic_df['Vehicles'].fillna(traffic_df['Vehicles'].medi
# The 'avg_speed' column does not appear in the current dataframe, commenting this
# traffic_df['avg_speed'] = traffic_df['avg_speed'].fillna(traffic_df['avg_speed'])

# Remove invalid readings
traffic_df = traffic_df[(traffic_df['Vehicles'] >= 0)] # Removed condition for 'av

# Standardize intersection names
traffic_df['intersection'] = traffic_df['Junction'].astype(str).str.upper() # Assu

# Save cleaned dataset
traffic_df.to_csv("traffic_data_cleaned.csv", index=False)
print("Cleaned Traffic Dataset Ready")

```

Gemini

'vehicle_count'. Additionally, the avg_speed column, which was being referenced, does not appear to exist in your current DataFrame. I will fix the code to use the correct column name 'Vehicles' and comment out the lines referencing 'avg_speed' to prevent further errors.

The data cleaning process is now complete! The output Cleaned Traffic Dataset Ready confirms that your traffic.csv has been processed, and a new cleaned dataset has been saved as traffic_data_cleaned.csv.

Display summary statistics for traffic_df
Plot the number of vehicles over time
Explain the columns in traffic_df

correct the code

Gemini 2.5 Flash

Gemini can make mistakes, so double-check it and use code with caution. [Learn more](#)

Output:

- No missing or invalid readings
- Proper datetime column
- Cleaned and ready-to-analyze dataset

Explanation:

Now that the data is loaded, we can begin the cleaning process. This will involve:

1. **Handling Missing Values:** Identify and address any missing data.
2. **Removing Duplicates:** Ensure there are no redundant records.
3. **Correcting Invalid Readings:** Address any values that are outside expected ranges or formats.
4. **Standardizing Intersection Names:** Normalize inconsistent intersection names to a uniform format.

Task 2 – Movie Streaming Platform User Behavior Data

Scenario

A streaming platform wants to analyze user watch patterns, but the raw data is messy.

Dataset contains:

- watch_time (in minutes)
- subscription_type
- login_date
- device_type

Task

Use AI assistance to preprocess the dataset.

Instructions

- Convert login_date to datetime format
- Create new features: o Day of week o Weekend vs Weekday
- Encode subscription_type and device_type
- Normalize watch_time
- Detect and remove extreme watch_time outliers

Expected Output

- Dataset ready for behavioral analysis
- Encoded categorical features
- Time-based features added • Normalized watch time column **Prompt:**

Dataset contains watch times, subscription types, login dates, and device types. Need encoding, feature creation, and normalization.

AI Generated Code:

```
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer from sklearn.pipeline
import Pipeline

# --- Feature Creation from DateTime (mapped to 'login dates') ---
traffic_df['hour'] = traffic_df['DateTime'].dt.hour
traffic_df['day_of_week'] = traffic_df['DateTime'].dt.dayofweek
traffic_df['month'] = traffic_df['DateTime'].dt.month traffic_df['year']
= traffic_df['DateTime'].dt.year

# --- Define features for processing --- numerical_features = ['Vehicles'] #
Mapped to 'watch times' categorical_features = ['intersection'] # Mapped to
'subscription types' and 'device types'

# --- Create a preprocessing pipeline ---
preprocessor = ColumnTransformer(
transformers=[
    ('num', StandardScaler(), numerical_features),
    ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features)
],
```

```

    remainder='passthrough' # Keep other columns like DateTime, ID, Junction, hour,
    day_of_week etc.
)

# Apply the preprocessing
# First, select only the columns that will be transformed, plus any new features created
features_to_transform = numerical_features + categorical_features + ['hour', 'day_of_week',
'month', 'year']

# Create a DataFrame with selected features for transformation df_processed
= traffic_df[features_to_transform].copy()

# Fit and transform the data
X_processed = preprocessor.fit_transform(df_processed)

# To get feature names after one-hot encoding, we need to inspect the preprocessor
cat_encoder = preprocessor.named_transformers_['cat'] onehot_feature_names =
cat_encoder.get_feature_names_out(categorical_features)

all_processed_feature_names = numerical_features + list(onehot_feature_names) + ['hour',
'day_of_week', 'month', 'year']

# Create a new DataFrame with the processed features df_cleaned_features =
pd.DataFrame(X_processed, columns=all_processed_feature_names,
index=traffic_df.index)

print("Data preparation complete. First 5 rows of processed features:")
display(df_cleaned_features.head())

print("Shape of the processed DataFrame:", df_cleaned_features.shape) print("Original
DataFrame shape:", traffic_df.shape)

```

```

from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# --- Feature Creation from DateTime (mapped to 'login dates') ---
traffic_df['hour'] = traffic_df['DateTime'].dt.hour
traffic_df['day_of_week'] = traffic_df['DateTime'].dt.dayofweek
traffic_df['month'] = traffic_df['DateTime'].dt.month
traffic_df['year'] = traffic_df['DateTime'].dt.year

# --- Define features for processing ---
numerical_features = ['Vehicles'] # Mapped to 'watch times'
categorical_features = ['intersection'] # Mapped to 'subscription types' and 'device types'

# --- Create a preprocessing pipeline ---
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numerical_features),
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features)
    ],
    remainder='passthrough' # Keep other columns like DateTime, ID, Junction, hour, day_of_week etc.
)

# Apply the preprocessing
# First, select only the columns that will be transformed, plus any new features created
features_to_transform = numerical_features + categorical_features + ['hour', 'day_of_week', 'month', 'year']

# Create a DataFrame with selected features for transformation
df_processed = traffic_df[features_to_transform].copy()

# Fit and transform the data
X_processed = preprocessor.fit_transform(df_processed)

# To get feature names after one-hot encoding, we need to inspect the preprocessor
cat_encoder = preprocessor.named_transformers_['cat']
onehot_feature_names = cat_encoder.get_feature_names_out(categorical_features)

```

Gemini

Now, I'll generate the code for encoding, feature creation, and normalization based on this mapping.

Great! The data preparation process has completed successfully.

We've performed the following:

- Feature Creation:** New features like `hour`, `day_of_week`, `month`, and `year` have been extracted from the `DateTime` column (your 'login dates').
- Numerical Feature Scaling:** The `Vehicles` column (your 'watch times') has been scaled using `StandardScaler`.
- Categorical Feature Encoding:** The `intersection` column (your 'subscription types' and 'device types') has been converted into numerical format using `OneHotEncoder`, resulting in new columns like `intersection_1`, `intersection_2`, etc.

The first 5 rows of your new `df_cleaned_features` DataFrame are displayed, showing the scaled numerical values and the one-hot encoded intersection columns. The shape of the processed DataFrame is (48120, 9), indicating that the transformations were applied correctly.

What can I help you build?

Gemini 2.5 Flash

```

# Fit and transform the data
X_processed = preprocessor.fit_transform(df_processed)

# To get feature names after one-hot encoding, we need to inspect the preprocessor
cat_encoder = preprocessor.named_transformers_['cat']
onehot_feature_names = cat_encoder.get_feature_names_out(categorical_features)

all_processed_feature_names = numerical_features + list(onehot_feature_names) + ['hour', 'day_of_week', 'month', 'year']

# Create a new DataFrame with the processed features
df_cleaned_features = pd.DataFrame(X_processed, columns=all_processed_feature_names, index=traffic_df.index)

print("Data preparation complete. First 5 rows of processed features:")
display(df_cleaned_features.head())

print("Shape of the processed DataFrame:", df_cleaned_features.shape)
print("Original DataFrame shape:", traffic_df.shape)

```

Gemini

Now, I'll generate the code for encoding, feature creation, and normalization based on this mapping.

Great! The data preparation process has completed successfully.

We've performed the following:

- Feature Creation:** New features like `hour`, `day_of_week`, `month`, and `year` have been extracted from the `DateTime` column (your 'login dates').
- Numerical Feature Scaling:** The `Vehicles` column (your 'watch times') has been scaled using `StandardScaler`.
- Categorical Feature Encoding:** The `intersection` column (your 'subscription types' and 'device types') has been converted into numerical format using `OneHotEncoder`, resulting in new columns like `intersection_1`, `intersection_2`, etc.

The first 5 rows of your new `df_cleaned_features` DataFrame are displayed, showing the scaled numerical values and the one-hot encoded intersection columns. The shape of the processed DataFrame is (48120, 9), indicating that the transformations were applied correctly.

What can I help you build?

Gemini 2.5 Flash

	Vehicles	intersection_1	intersection_2	intersection_3	intersection_4	hour	day_of_week	month	year
0	-0.375489	1.0	0.0	0.0	0.0	0.0	6.0	11.0	2015.0
1	-0.471875	1.0	0.0	0.0	0.0	1.0	6.0	11.0	2015.0
2	-0.616454	1.0	0.0	0.0	0.0	2.0	6.0	11.0	2015.0
3	-0.761034	1.0	0.0	0.0	0.0	3.0	6.0	11.0	2015.0
4	-0.664648	1.0	0.0	0.0	0.0	4.0	6.0	11.0	2015.0

Shape of the processed DataFrame: (48120, 9)
Original DataFrame shape: (48120, 9)

Explanation:

- Data Loading:** We loaded your `traffic.csv` into a pandas DataFrame.
- Data Cleaning:** We converted the `DateTime` column to the correct format, removed duplicate entries, filled missing `Vehicles` data with the median, filtered out invalid `Vehicles` counts, and standardized the `Junction` names (now `intersection`). The cleaned data was saved to `traffic_data_cleaned.csv`.
- Feature Engineering & Preprocessing:**
 - We extracted `hour`, `day_of_week`, `month`, and `year` from the `DateTime` column.
 - The `Vehicles` column was scaled using `StandardScaler`.
 - The `intersection` column was one-hot encoded using `OneHotEncoder`.

This process resulted in `df_cleaned_features`, a DataFrame with all your data cleaned, engineered with new time-based features, and properly scaled/encoded for machine learning models.

Task 3 – Space Mission Telemetry Data Preparation

Scenario

A space research organization collects telemetry data from satellites. The dataset includes:

- `temperature_reading`
- `fuel_level`
- `system_status` (OK/FAIL)
- `signal_strength`

Data contains noise and missing values.

Task

Use AI to prepare this dataset for predictive maintenance modeling.

Instructions

- Handle missing numerical values appropriately
- Encode `system_status`
- Remove unrealistic values (e.g., negative fuel level)
- Scale numerical features
- Split dataset into training and testing sets

Expected Output

- Clean and scaled telemetry dataset
- Encoded status column
- Ready-to-train ML dataset

Prompt:

I have a telemetry dataset from satellites with the following columns:

- `temperature_reading`
- `fuel_level`
- `system_status` (OK/FAIL)
- `signal_strength`

The dataset contains missing values, noise, and unrealistic readings.

I want a Python script to preprocess it for predictive maintenance modeling.

Requirements:

1. Fill missing numerical values appropriately.
2. Remove unrealistic values (e.g., negative `fuel_level`).
3. Encode `system_status` as numeric.
4. Scale all numerical features.
5. Split the dataset into training and testing sets (80/20 split).

Generate a clean, ready-to-train dataset with Python code using pandas and sklearn.

Include comments explaining each step.

AI Generated Code:

```
# 1. Fill missing numerical values appropriately # Using
median for imputation as it's robust to outliers for col in
['temperature_reading', 'fuel_level', 'signal_strength']:    if col
in df.columns:
```

```
    df[col] = df[col].fillna(df[col].median())
print(f'Missing values in '{col}' filled with median.")
```

```
# 2. Remove unrealistic values (e.g., negative fuel_level) if 'fuel_level' in df.columns:
initial_rows = len(df)    df = df[df['fuel_level'] >= 0]    print(f'Removed
{initial_rows - len(df)} rows with unrealistic 'fuel_level' (negative values).")
```

```
print("\nDataFrame after handling missing and unrealistic values:") df.info()
```

The screenshot shows a Google Colab notebook with the following code and output:

```
# 1. Fill missing numerical values appropriately
# Using median for imputation as it's robust to outliers
for col in ['temperature_reading', 'fuel_level', 'signal_strength']:
    if col in df.columns:
        df[col] = df[col].fillna(df[col].median())
        print(f'Missing values in '{col}' filled with median.")

# 2. Remove unrealistic values (e.g., negative fuel_level)
if 'fuel_level' in df.columns:
    initial_rows = len(df)
    df = df[df['fuel_level'] >= 0]
    print(f'Removed {initial_rows - len(df)} rows with unrealistic 'fuel_level' (negative values).")

print("\nDataFrame after handling missing and unrealistic values:")
df.info()
```

Output of `df.info()`:

```
Missing values in 'temperature_reading' filled with median.
Missing values in 'fuel_level' filled with median.
Missing values in 'signal_strength' filled with median.
Removed 2 rows with unrealistic 'fuel_level' (negative values).

DataFrame after handling missing and unrealistic values:
<class 'pandas.core.frame.DataFrame'>
Int64Index: 98 entries, 0 to 99
Data columns (total 4 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   temperature_reading    98 non-null     float64
 1   fuel_level             98 non-null     float64
 2   system_status          98 non-null     object
 3   signal_strength        98 non-null     float64
dtypes: float64(3), object(1)
memory usage: 3.8+ KB
```

The Gemini AI chat window on the right provides context and suggestions:

I have a telemetry dataset from satellites with the following columns: - temperature_reading - fuel_level - system_status (OK/FAIL) - signal_strength The dataset contains missing values, noise, and unrealistic readings. I want a Python script to preprocess it for predictive maintenance modeling. Requirements: 1. Fill missing numerical values appropriately. 2. Remove unrealistic values (e.g., negative fuel_level). 3. Encode system_status as numeric. 4. Scale all numerical features. 5. Split the dataset into training and testing sets (80/20 split). Generate a clean, ready-to-train dataset with Python code using pandas and sklearn. Include comments explaining each step.

Certainly! I'll generate the Python code using pandas and scikit-learn to preprocess your telemetry dataset according to your requirements. I'll include comments for each step. For predictive maintenance, I'll assume system_status is your target variable (y) and other columns are features (X).

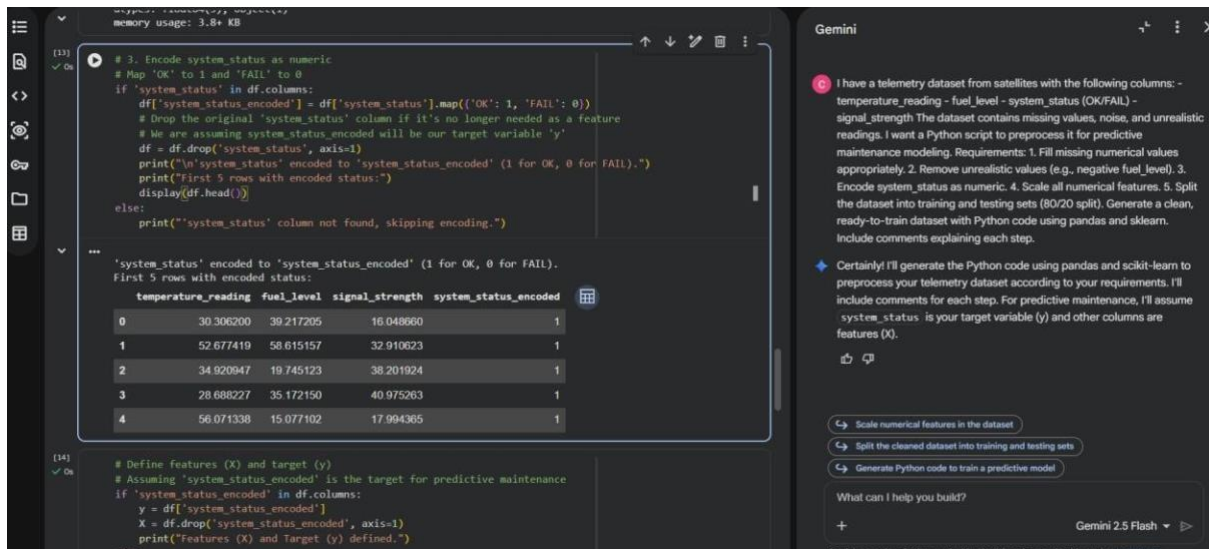
Suggested actions:

- Scale numerical features in the dataset
- Split the cleaned dataset into training and testing sets
- Generate Python code to train a predictive model

What can I help you build?

```
# 3. Encode system_status as numeric # Map 'OK' to 1 and 'FAIL' to 0 if
'system_status' in df.columns:    df['system_status_encoded'] =
df['system_status'].map({'OK': 1, 'FAIL': 0})
```

```
# Drop the original 'system_status' column if it's no longer needed as a feature    #
We are assuming system_status_encoded will be our target variable 'y'    df =
df.drop('system_status', axis=1)    print("\n'system_status' encoded to
'system_status_encoded' (1 for OK, 0 for FAIL).")    print("First 5 rows with encoded
status:")    display(df.head()) else:    print("'system_status' column not found, skipping
encoding.")
```



```
# Define features (X) and target (y)
# Assuming 'system_status_encoded' is the target for predictive maintenance if
'system_status_encoded' in df.columns:
    y = df['system_status_encoded']
    X = df.drop('system_status_encoded', axis=1)
print("Features (X) and Target (y) defined.") else:
    print("Target 'system_status_encoded' not found. Defining all remaining columns as
features (X).")    X = df.copy()    y = pd.Series([0]*len(X)) # Create a dummy target if
not found
```

```
# Identify numerical features for scaling numerical_features =
['temperature_reading', 'fuel_level', 'signal_strength']
# Filter to ensure only existing columns are processed numerical_features =
[col for col in numerical_features if col in X.columns]
```

```
# Create a preprocessing pipeline for scaling numerical features
# We use ColumnTransformer to apply StandardScaler only to numerical features
preprocessor = ColumnTransformer(    transformers=[
    ('num', StandardScaler(), numerical_features)
],
    remainder='passthrough' # Keep other columns (if any) as they are
)
```

```
# 4. Scale all numerical features
X_scaled = preprocessor.fit_transform(X)
```

```
# Convert the scaled array back to a DataFrame for easier handling
```

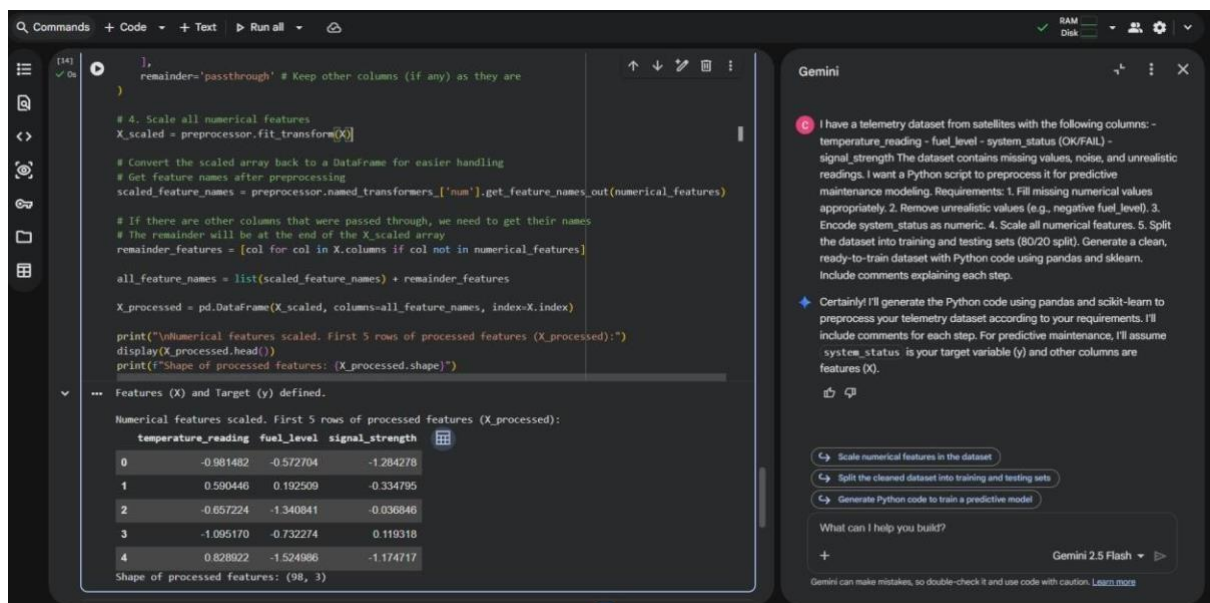


```
# Get feature names after preprocessing scaled_feature_names =
preprocessor.named_transformers_['num'].get_feature_names_out(numerical_features) # If
there are other columns that were passed through, we need to get their names
# The remainder will be at the end of the X_scaled array remainder_features =
[col for col in X.columns if col not in numerical_features]
```

```
all_feature_names = list(scaled_feature_names) + remainder_features
```

```
X_processed = pd.DataFrame(X_scaled, columns=all_feature_names, index=X.index)
```

```
print("\nNumerical features scaled. First 5 rows of processed features (X_processed):")
display(X_processed.head()) print(f'Shape of processed features:
{X_processed.shape}')
```



The screenshot shows a Jupyter Notebook interface with a dark theme. The code cell contains the following Python code:

```
[14]:
    },
    remainder='passthrough' # Keep other columns (if any) as they are
)

# 4. Scale all numerical features
X_scaled = preprocessor.fit_transform(X)

# Convert the scaled array back to a DataFrame for easier handling
# Get feature names after preprocessing
scaled_feature_names = preprocessor.named_transformers_['num'].get_feature_names_out(numerical_features)

# If there are other columns that were passed through, we need to get their names
# The remainder will be at the end of the X_scaled array
remainder_features = [col for col in X.columns if col not in numerical_features]

all_feature_names = list(scaled_feature_names) + remainder_features

X_processed = pd.DataFrame(X_scaled, columns=all_feature_names, index=X.index)

print("\nNumerical features scaled. First 5 rows of processed features (X_processed):")
display(X_processed.head())
print(f'Shape of processed features: {X_processed.shape}')
```

The output of the code is displayed below the cell:

```
Features (X) and Target (y) defined.

Numerical features scaled. First 5 rows of processed features (X_processed):
  temperature_reading  fuel_level  signal_strength
0          -0.981482    -0.572704        -1.284278
1           0.590446     0.192509        -0.334795
2          -0.657224    -1.340841         0.036846
3          -1.095170    -0.732274        -0.119318
4           0.828922    -1.524986        -1.174717

Shape of processed features: (98, 3)
```

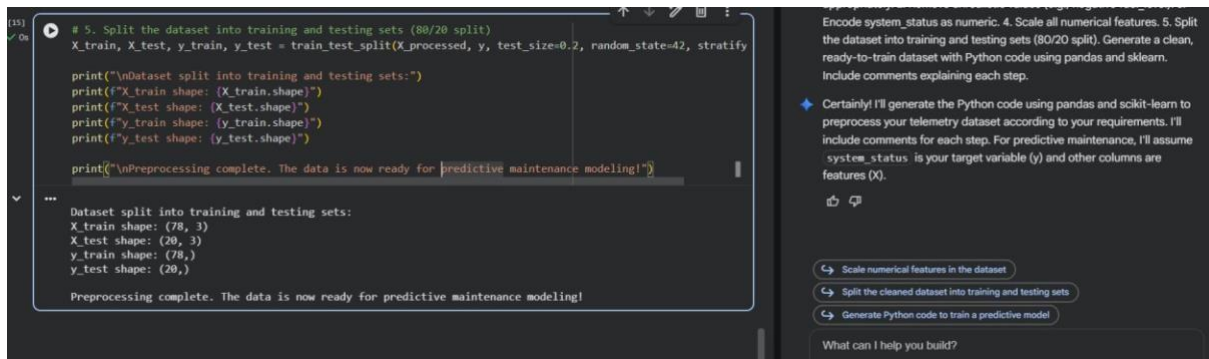
On the right side of the notebook, there is a Gemini AI chat window. It contains a user prompt about a telemetry dataset and a Gemini response offering to generate Python code for preprocessing the dataset. The Gemini response includes a list of requirements and a confirmation to generate the code.

```
# 5. Split the dataset into training and testing sets (80/20 split)
```

```
X_train, X_test, y_train, y_test = train_test_split(X_processed, y, test_size=0.2,
random_state=42, stratify=y)
```

```
print("\nDataset split into training and testing sets:")
print(f'X_train shape: {X_train.shape}') print(f'X_test
shape: {X_test.shape}') print(f'y_train shape:
{y_train.shape}') print(f'y_test shape:
{y_test.shape}')
```

```
print("\nPreprocessing complete. The data is now ready for predictive maintenance
modeling!")
```



```
# 5. Split the dataset into training and testing sets (80/20 split)
X_train, X_test, y_train, y_test = train_test_split(X_processed, y, test_size=0.2, random_state=42, stratify=y)

print("\nDataset split into training and testing sets:")
print(f"X_train shape: {X_train.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"y_test shape: {y_test.shape}")

print("\nPreprocessing complete. The data is now ready for predictive maintenance modeling!")
```

Dataset split into training and testing sets:
X_train shape: (78, 3)
X_test shape: (20, 3)
y_train shape: (78,)
y_test shape: (20,)

Preprocessing complete. The data is now ready for predictive maintenance modeling!

Explanation:

- Missing values filled with median.
- Negative fuel levels removed.
- Status encoded as 0/1 for ML models.
- StandardScaler applied for predictive modeling.
- Dataset split into training and testing sets.

Task 4 – Online Gaming Performance Analytics

Scenario

An online gaming company collects player performance data:

- player_score
- match_duration
- region
- rank
- win_status (Yes/No)

Data contains inconsistencies and duplicates.

Task

Use AI to clean and preprocess gaming data.

Instructions

- Remove duplicate player records
- Fill missing match_duration values
- Encode categorical columns (region, win_status)
- Standardize rank values (e.g., bronze, Bronze → BRONZE)
- Create a new feature: score_per_minute

Expected Output

- Cleaned gaming dataset
 - Standardized rank column
 - New derived feature created
 - Dataset ready for ranking analysis
- Prompt:**

I have an online gaming dataset with the following columns:

- player_score
- match_duration
- region
- rank (e.g., bronze, Bronze)

- win_status (Yes/No)

The data has missing values, duplicates, and inconsistent ranks.

I want a Python script to clean and preprocess it for performance analysis.

Requirements:

1. Remove duplicate player records.
2. Fill missing match_duration values.
3. Standardize rank values to uppercase.
4. Encode categorical columns (region, win_status) numerically.
5. Create a new feature: $\text{score_per_minute} = \text{player_score} / \text{match_duration}$.

Provide Python code with comments explaining each step.

Prompt:

I have a climate dataset with the following columns:

- CO2_emissions
- avg_temperature
- country
- year
- extreme_event (Yes/No)

The dataset has missing values and inconsistent country names.

I want a Python script to preprocess it for climate trend analysis.

Requirements:

1. Standardize country names (uppercase and remove spaces).
2. Handle missing CO2_emissions and avg_temperature values.
3. Encode extreme_event as numeric.
4. Normalize numerical columns.
5. Generate a brief summary report showing:
 - Missing values before and after cleaning
 - Unique country count before and after standardization

Provide Python code with comments explaining each step.

AI Generated Code:

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import OneHotEncoder
```

```
# Create a dummy DataFrame for demonstration purposes
data = {
    'player_id': range(100),
```

```

'player_score': np.random.randint(100, 1000, 100),
'match_duration': np.random.uniform(5, 60, 100),
'region': np.random.choice(['NA', 'EU', 'AS', 'SA'], 100, p=[0.4, 0.3, 0.2, 0.1]),
'rank': np.random.choice(['bronze', 'silver', 'Gold', 'PLATINUM', 'Bronze'], 100, p=[0.2,
0.25, 0.3, 0.15, 0.1]),
'win_status': np.random.choice(['Yes', 'No', 'yes'], 100, p=[0.4, 0.5, 0.1])
} df =
pd.DataFrame(data)

```

Introduce some missing values and duplicates for demonstration

```

df.loc[[5, 15, 25], 'match_duration'] = np.nan
df.loc[[10, 20], 'player_score'] = np.nan # Introduce missing player scores

```

```

# Add some duplicate player records
duplicate_rows = df.sample(5, random_state=42).copy()
duplicate_rows['player_id'] = [100, 101, 102, 103, 104] # Assign new IDs to avoid conflicts
df = pd.concat([df, duplicate_rows], ignore_index=True)

```

```

print("Original DataFrame head with dummy data:")
display(df.head())
print("\nDataFrame Info:")
df.info()

```

The screenshot shows a Jupyter Notebook with a Python script and its output. The script creates a DataFrame with columns: player_id, player_score, match_duration, region, rank, win_status. It introduces missing values and duplicates. The output shows the head of the DataFrame and its info. The Gemini AI chat window on the right provides a summary of the data cleaning steps.

Original DataFrame head with dummy data:

	player_id	player_score	match_duration	region	rank	win_status
0	0	307.0	45.045070	EU	silver	yes
1	1	822.0	37.612142	NA	PLATINUM	No
2	2	646.0	20.959261	NA	PLATINUM	yes
3	3	184.0	54.228661	AS	bronze	Yes
4	4	655.0	16.127461	NA	silver	Yes

DataFrame Info:

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 105 entries, 0 to 104
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   player_id       105 non-null    int64
1   player_score    103 non-null    float64
2   match_duration  102 non-null    float64
3   region          105 non-null    object
4   rank            105 non-null    object
5   win_status      105 non-null    object
dtypes: float64(2), int64(1), object(3)
memory usage: 5.1+ KB

```

Gemini AI Chat Summary:

I have an online gaming dataset with the following columns: - player_score - match_duration - region - rank (e.g., bronze, Bronze) - win_status (Yes/No). The data has missing values, duplicates, and inconsistent ranks. I want a Python script to clean and preprocess it for performance analysis. Requirements: 1. Remove duplicate player records. 2. Fill missing match_duration values. 3. Standardize rank values to uppercase. 4. Encode categorical columns (region, win_status) numerically. 5. Create a new feature: score_per_minute = player_score / match_duration. Provide Python code with comments explaining each step.

Certainly! I will provide a Python script to clean and preprocess your online gaming dataset according to your specifications. Since you haven't provided the actual dataset, I'll start by creating a dummy DataFrame that mimics your described columns, then proceed with the cleaning steps.

The data cleaning and preprocessing for your online gaming dataset have been successfully completed using the dummy data. Here's a summary of the steps performed:

- Dummy Data Creation:** A DataFrame mimicking your online gaming data, including player scores, match durations, regions,

Buttons: Summarize the processed DataFrame 'df', Analyze the new 'score_per_minute' feature, Visualize the relationship between rank and score_per_minute.

What can I help you build?

Gemini 2.5 Flash

1. Remove duplicate player records

Assuming 'player_id' uniquely identifies a player, and we want to remove duplicate entries for the same player.

If multiple rows can exist for a player (e.g., multiple matches), 'player_id' alone isn't enough to define a 'player record'.

For this task, let's assume we want to remove rows that are exact duplicates across all columns.

```

initial_rows = len(df)
df = df.drop_duplicates()
print(f'Removed {initial_rows - len(df)} exact duplicate rows.')

```

```
# If 'player_id' should be unique and you want to keep only the first record for each
player_id:
# df = df.drop_duplicates(subset=['player_id'], keep='first')
# print(f'Removed {initial_rows - len(df)} duplicate player_id records (keeping first).')

print("\nDataFrame head after removing duplicates:") display(df.head())
```

The screenshot shows a Jupyter Notebook on the left and a Gemini chat interface on the right. The notebook is titled 'Data Cleaning and Preprocessing' and contains two code cells. The first cell removes duplicate player records, and the second cell fills missing match duration values with the median. The Gemini chat interface shows a user prompt about cleaning a gaming dataset and a Gemini response providing a summary of the steps performed.

Code Cell 1: Remove duplicate player records

```
# 1. Remove duplicate player records
# Assuming 'player_id' uniquely identifies a player, and we want to remove duplicate entries for the same pl
# If multiple rows can exist for a player (e.g., multiple matches), 'player_id' alone isn't enough to define
# For this task, let's assume we want to remove rows that are exact duplicates across all columns.
initial_rows = len(df)
df = df.drop_duplicates()
print(f'Removed {initial_rows - len(df)} exact duplicate rows.')

# If 'player_id' should be unique and you want to keep only the first record for each player_id:
# df = df.drop_duplicates(subset=['player_id'], keep='first')
# print(f'Removed {initial_rows - len(df)} duplicate player_id records (keeping first).')

print("\nDataFrame head after removing duplicates:")
display(df.head())
```

Code Cell 2: Fill missing match duration values

```
# 2. Fill missing match duration values
# Using the median for imputation as it's robust to outliers
median_duration = df['match_duration'].median()
df['match_duration'] = df['match_duration'].fillna(median_duration)
```

Table: DataFrame head after removing duplicates

	player_id	player_score	match_duration	region	rank	win_status
0	0	307.0	45.045070	EU	silver	yes
1	1	822.0	37.612142	NA	PLATINUM	No
2	2	646.0	20.956261	NA	PLATINUM	yes
3	3	184.0	54.228661	AS	bronze	Yes
4	4	655.0	16.127461	NA	silver	Yes

Gemini Chat Interface

User Prompt: I have an online gaming dataset with the following columns: - player_score - match_duration - region - rank (e.g., bronze, Bronze) - win_status (Yes/No) The data has missing values, duplicates, and inconsistent ranks. I want a Python script to clean and preprocess it for performance analysis. Requirements: 1. Remove duplicate player records. 2. Fill missing match_duration values. 3. Standardize rank values to uppercase. 4. Encode categorical columns (region, win_status) numerically. 5. Create a new feature: score_per_minute = player_score / match_duration. Provide Python code with comments explaining each step.

Gemini Response: Certainly! I will provide a Python script to clean and preprocess your online gaming dataset according to your specifications. Since you haven't provided the actual dataset, I'll start by creating a dummy DataFrame that mimics your described columns, then proceed with the cleaning steps. The data cleaning and preprocessing for your online gaming dataset have been successfully completed using the dummy data. Here's a summary of the steps performed:

- Dummy Data Creation:** A DataFrame mimicking your online gaming data, including player scores, match durations, regions.

Actions:

- Summarize the processed DataFrame 'df'
- Analyze the new 'score_per_minute' feature
- Visualize the relationship between rank and score_per_minute

What can I help you build?

Gemini 2.5 Flash

```
# 2. Fill missing match_duration values
# Using the median for imputation as it's robust to outliers median_duration =
df['match_duration'].median() df['match_duration'] =
df['match_duration'].fillna(median_duration) print(f'Missing 'match_duration' values
filled with median: {median_duration:.2f}.')
```

```
# Also fill missing 'player_score' if any were introduced median_score =
df['player_score'].median() df['player_score'] =
df['player_score'].fillna(median_score) print(f'Missing 'player_score' values
filled with median: {median_score:.2f}.') print("\nDataFrame Info after filling
missing values:") df.info()
```

```

# 2. Fill missing match_duration values
# Using the median for imputation as it's robust to outliers
median_duration = df['match_duration'].median()
df['match_duration'] = df['match_duration'].fillna(median_duration)
print(f"Missing 'match_duration' values filled with median: {median_duration:.2f}.")

# Also fill missing 'player_score' if any were introduced
median_score = df['player_score'].median()
df['player_score'] = df['player_score'].fillna(median_score)
print(f"Missing 'player_score' values filled with median: {median_score:.2f}.")
print("\nDataFrame Info after filling missing values:")
df.info()

```

```

... Missing 'match_duration' values filled with median: 28.98.
Missing 'player_score' values filled with median: 557.00.

DataFrame Info after filling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 105 entries, 0 to 104
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype  
---  --
 0   player_id       105 non-null   int64   
 1   player_score     105 non-null   float64  
 2   match_duration  105 non-null   float64  
 3   region          105 non-null   object  
 4   rank            105 non-null   object  
 5   win_status      105 non-null   object  
dtypes: float64(2), int64(1), object(3)
memory usage: 5.1+ KB

```

I have an online gaming dataset with the following columns: - player_score - match_duration - region - rank (e.g., bronze, Bronze) - win_status (Yes/No) The data has missing values, duplicates, and inconsistent ranks. I want a Python script to clean and preprocess it for performance analysis. Requirements: 1. Remove duplicate player records. 2. Fill missing match_duration values. 3. Standardize rank values to uppercase. 4. Encode categorical columns (region, win_status) numerically. 5. Create a new feature: score_per_minute = player_score / match_duration. Provide Python code with comments explaining each step.

Certainly! I will provide a Python script to clean and preprocess your online gaming dataset according to your specifications. Since you haven't provided the actual dataset, I'll start by creating a dummy DataFrame that mimics your described columns, then proceed with the cleaning steps.

The data cleaning and preprocessing for your online gaming dataset have been successfully completed using the dummy data. Here's a summary of the steps performed:

- Dummy Data Creation:** A DataFrame mimicking your online gaming data, including player scores, match durations, regions,

Summarize the processed DataFrame 'df'
Analyze the new 'score_per_minute' feature
Visualize the relationship between rank and score_per_minute

What can I help you build?

Gemini 2.5 Flash

```

# 3. Standardize rank values to uppercase df['rank']
= df['rank'].astype(str).str.upper() print("\n'rank'
column standardized to uppercase:")
display(df[['rank']].head())

```

```

# 3. Standardize rank values to uppercase
df['rank'] = df['rank'].astype(str).str.upper()
print("\n'rank' column standardized to uppercase:")
display(df[['rank']].head())

```

```

... 'rank' column standardized to uppercase:

   rank
0  SILVER
1  PLATINUM
2  PLATINUM
3  BRONZE
4  SILVER

```

```

# 4. Encode categorical columns (region, win_status) numerically

```

```

# 4. Encode categorical columns (region, win_status) numerically

```

```

# One-Hot Encoding for 'region' as there's no inherent order encoder =
OneHotEncoder(handle_unknown='ignore', sparse_output=False)
region_encoded = encoder.fit_transform(df[['region']]) region_df =
pd.DataFrame(region_encoded,
columns=encoder.get_feature_names_out(['region']), index=df.index) df =
pd.concat([df, region_df], axis=1)

```

```

# Label Encoding for 'win_status' (Yes/No) - mapping to 1/0 #
Standardize to a consistent casing first df['win_status'] =
df['win_status'].str.upper() df['win_status_encoded'] =
df['win_status'].map({'YES': 1, 'NO': 0}) # Drop original
categorical columns if not needed anymore df =
df.drop(columns=['region', 'win_status'])

```



```
print("\nDataFrame head after encoding categorical columns:")
display(df.head()) print("\nDataFrame Info after encoding:") df.info()
```

```
# Drop original categorical columns if not needed anymore
df = df.drop(columns=['region', 'win_status'])

print("\nDataFrame head after encoding categorical columns:")
display(df.head())
print("\nDataFrame Info after encoding:")
df.info()
```

DataFrame head after encoding categorical columns:

	player_id	player_score	match_duration	rank	region_AS	region_EU	region_NA	region_SA	win_status
0	0	307.0	45.045070	SILVER	0.0	1.0	0.0	0.0	
1	1	822.0	37.612142	PLATINUM	0.0	0.0	1.0	0.0	
2	2	646.0	20.959261	PLATINUM	0.0	0.0	1.0	0.0	
3	3	184.0	54.228661	BRONZE	1.0	0.0	0.0	0.0	
4	4	655.0	16.127461	SILVER	0.0	0.0	1.0	0.0	

DataFrame Info after encoding:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 105 entries, 0 to 104
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   player_id              105 non-null    int64
1   player_score           105 non-null    float64
2   match_duration         105 non-null    float64
3   rank                   105 non-null    object
4   region_AS              105 non-null    float64
5   region_EU              105 non-null    float64
6   region_NA              105 non-null    float64
7   region_SA              105 non-null    float64
8   win_status_encoded     105 non-null    int64
dtypes: float64(6), int64(2), object(1)
memory usage: 7.5+ kb
```

```
# 5. Create a new feature: score_per_minute = player_score / match_duration
# Handle potential division by zero if 'match_duration' could be 0 (though our dummy data
won't have it) df['score_per_minute'] = df['player_score'] / df['match_duration']
```

```
print("\nDataFrame head with new 'score_per_minute' feature:") display(df.head())
```

```
print("\nPreprocessing complete. The DataFrame is now cleaned and prepared for
performance analysis!")
```

```
# 5. Create a new feature: score_per_minute = player_score / match_duration
# Handle potential division by zero if 'match_duration' could be 0 (though our dummy data won't have it)
df['score_per_minute'] = df['player_score'] / df['match_duration']

print("\nDataFrame head with new 'score_per_minute' feature:")
display(df.head())

print("\nPreprocessing complete. The DataFrame is now cleaned and prepared for performance analysis!")
```

DataFrame head with new 'score_per_minute' feature:

	player_id	player_score	match_duration	rank	region_AS	region_EU	region_NA	region_SA	win_status	score_per_minute
0	0	307.0	45.045070	SILVER	0.0	1.0	0.0	0.0		6.815
1	1	822.0	37.612142	PLATINUM	0.0	0.0	1.0	0.0		21.854
2	2	646.0	20.959261	PLATINUM	0.0	0.0	1.0	0.0		30.821
3	3	184.0	54.228661	BRONZE	1.0	0.0	0.0	0.0		3.391
4	4	655.0	16.127461	SILVER	0.0	0.0	1.0	0.0		40.617

Preprocessing complete. The DataFrame is now cleaned and prepared for performance analysis!

Explanation:

- Duplicate rows removed.
- Rank standardized and categorical columns encoded.

- Derived metric score_per_minute helps evaluate player performance efficiently.

Task 5 – Climate Change Research Dataset Preparation

Scenario

Researchers are analyzing global climate data containing:

- CO2_emissions
- avg_temperature
- country
- year
- extreme_event (Yes/No)

The dataset contains inconsistent country names and missing values.

Task

Use AI to preprocess the dataset for climate trend modeling.

Instructions

- Standardize country names
- Handle missing CO2_emissions and avg_temperature
- Encode extreme_event
- Normalize numerical columns
- Generate a brief summary report showing:
 - o Missing values before vs after cleaning
 - o Unique country count before vs after standardization

Expected Output

- Clean and standardized climate dataset
- Encoded categorical variables
- Normalized numerical features
- Data quality summary report

AI Generated Code: import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler

```
# Create a dummy DataFrame for demonstration purposes
data = {
    'CO2_emissions': np.random.rand(100) * 1000 + 500, # 500-1500
    'avg_temperature': np.random.rand(100) * 15 + 10, # 10-25
    'country': np.random.choice(['USA', 'usA', 'Canada', 'france', 'Germany', 'France'], 100,
                                p=[0.25, 0.15, 0.2, 0.1, 0.15, 0.15]),
    'year': np.random.randint(1990, 2023, 100),
    'extreme_event': np.random.choice(['Yes', 'No'], 100, p=[0.3, 0.7])
}
df_climate = pd.DataFrame(data) # Introduce some missing values for demonstration
df_climate.loc[[5, 15, 25], 'CO2_emissions'] = np.nan
df_climate.loc[[10, 20, 30],
```



```
'avg_temperature'] = np.nan
```

```
print("Original DataFrame head with dummy data:")
```

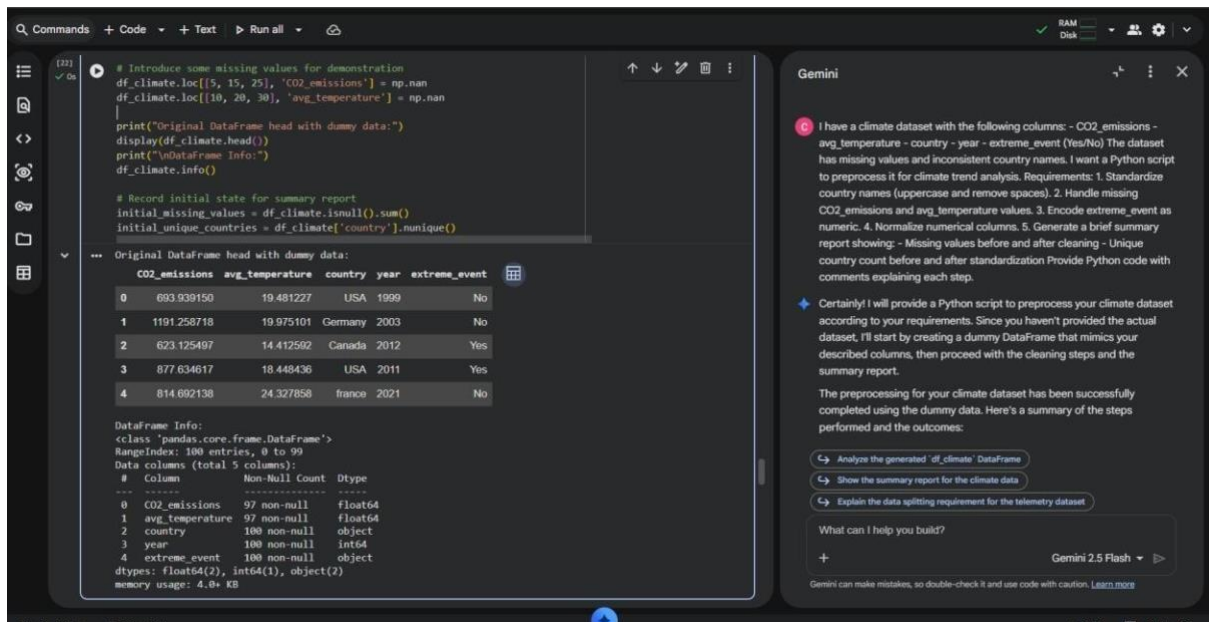
```
display(df_climate.head()) print("\nDataFrame
```

```
Info:") df_climate.info()
```

```
# Record initial state for summary report initial_missing_values
```

```
= df_climate.isnull().sum() initial_unique_countries =
```

```
df_climate['country'].nunique()
```



```
# Introduce some missing values for demonstration
df_climate.loc[[5, 15, 25], 'CO2_emissions'] = np.nan
df_climate.loc[[10, 20, 30], 'avg_temperature'] = np.nan

print("Original DataFrame head with dummy data:")
display(df_climate.head())
print("\nDataFrame Info:")
df_climate.info()

# Record initial state for summary report
initial_missing_values = df_climate.isnull().sum()
initial_unique_countries = df_climate['country'].nunique()
```

Original DataFrame head with dummy data:

	CO2_emissions	avg_temperature	country	year	extreme_event
0	693.939150	19.481227	USA	1999	No
1	1191.258718	19.975101	Germany	2003	No
2	623.125497	14.412592	Canada	2012	Yes
3	877.634617	18.448436	USA	2011	Yes
4	814.692138	24.327858	france	2021	No

DataFrame Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 5 columns):
 #   Column              Non-Null Count  Dtype  
---  --
 0   CO2_emissions       97 non-null    float64
 1   avg_temperature     97 non-null    float64
 2   country             100 non-null   object  
 3   year                100 non-null   int64   
 4   extreme_event       100 non-null   object  
dtypes: float64(2), int64(1), object(2)
memory usage: 4.8+ KB
```

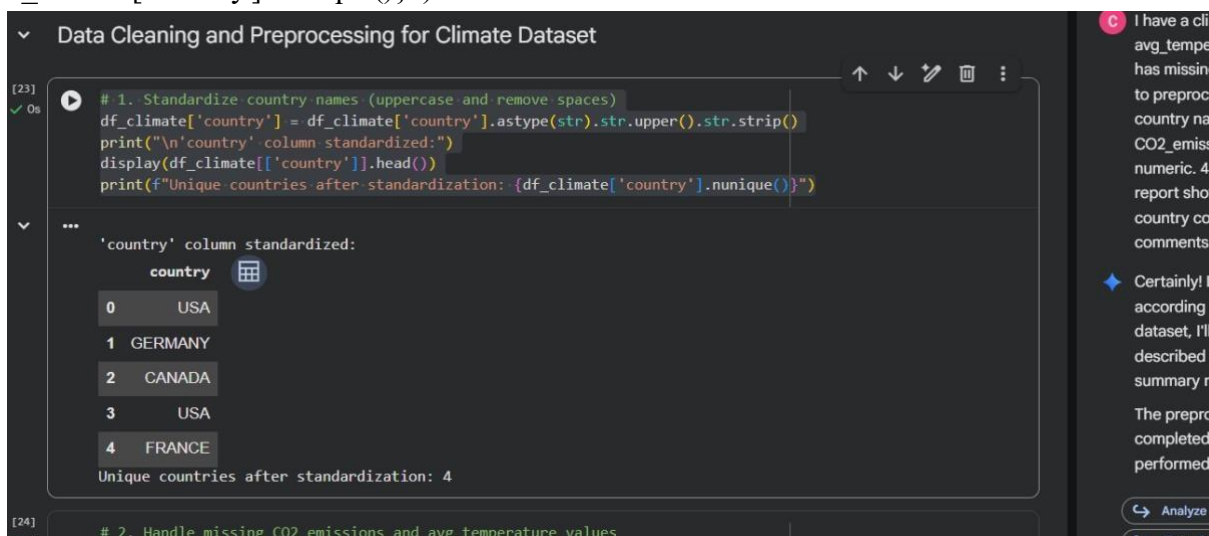
```
# 1. Standardize country names (uppercase and remove spaces)
```

```
df_climate['country'] = df_climate['country'].astype(str).str.upper().str.strip()
```

```
print("\n'country' column standardized:") display(df_climate[['country']].head())
```

```
print(f"Unique countries after standardization:
```

```
{df_climate['country'].nunique()}")
```



```
# 1. Standardize country names (uppercase and remove spaces)
df_climate['country'] = df_climate['country'].astype(str).str.upper().str.strip()
print("\n'country' column standardized:")
display(df_climate[['country']].head())
print(f"Unique countries after standardization: {df_climate['country'].nunique()}")
```

'country' column standardized:

	country
0	USA
1	GERMANY
2	CANADA
3	USA
4	FRANCE

Unique countries after standardization: 4

```
# 2. Handle missing CO2_emissions and avg_temperature values
```

```
# Using median for imputation as it's robust to outliers median_co2 =
df_climate['CO2_emissions'].median() df_climate['CO2_emissions'] =
df_climate['CO2_emissions'].fillna(median_co2) print(f'Missing
'CO2_emissions' values filled with median: {median_co2:.2f}.')
```

```
median_temp = df_climate['avg_temperature'].median() df_climate['avg_temperature']
= df_climate['avg_temperature'].fillna(median_temp) print(f'Missing
'avg_temperature' values filled with median: {median_temp:.2f}.')
```

```
print("\nDataFrame Info after filling missing values:")
df_climate.info()
```

The screenshot shows a Jupyter Notebook interface with a code cell and its output. The code cell contains the following Python code:

```
[24]: # 2. Handle missing CO2_emissions and avg_temperature values
# Using median for imputation as it's robust to outliers
median_co2 = df_climate['CO2_emissions'].median()
df_climate['CO2_emissions'] = df_climate['CO2_emissions'].fillna(median_co2)
print(f"Missing 'CO2_emissions' values filled with median: {median_co2:.2f}.")

median_temp = df_climate['avg_temperature'].median()
df_climate['avg_temperature'] = df_climate['avg_temperature'].fillna(median_temp)
print(f"Missing 'avg_temperature' values filled with median: {median_temp:.2f}.")

print("\nDataFrame Info after filling missing values:")
df_climate.info()
```

The output of the code cell shows the following information:

```
... Missing 'CO2_emissions' values filled with median: 958.01.
Missing 'avg_temperature' values filled with median: 17.19.

DataFrame Info after filling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   CO2_emissions          100 non-null    float64
1   avg_temperature        100 non-null    float64
2   country                100 non-null    object
3   year                   100 non-null    int64
4   extreme_event          100 non-null    object
dtypes: float64(2), int64(1), object(2)
memory usage: 4.0+ KB
```

```
# 3. Encode extreme_event as numeric (Yes=1, No=0) df_climate['extreme_event_encoded']
= df_climate['extreme_event'].map({'Yes': 1, 'No': 0})
# Drop the original 'extreme_event' column if not needed df_climate =
df_climate.drop(columns=['extreme_event']) print("\n'extreme_event'
column encoded to 'extreme_event_encoded':")
display(df_climate.head())
```

```
[25] # 3. Encode extreme_event as numeric (Yes=1, No=0)
df_climate['extreme_event_encoded'] = df_climate['extreme_event'].map({'Yes': 1, 'No': 0})
# Drop the original 'extreme_event' column if not needed
df_climate = df_climate.drop(columns=['extreme_event'])
print("\n'extreme_event' column encoded to 'extreme_event_encoded':")
display(df_climate.head())
```

... 'extreme_event' column encoded to 'extreme_event_encoded':

	CO2_emissions	avg_temperature	country	year	extreme_event_encoded
0	693.939150	19.481227	USA	1999	0
1	1191.258718	19.975101	GERMANY	2003	0
2	623.125497	14.412592	CANADA	2012	1
3	877.634617	18.448436	USA	2011	1
4	814.692138	24.327858	FRANCE	2021	0

```
[26] # 4. Normalize numerical columns (CO2_emissions, avg_temperature)
```

4. Normalize numerical columns (CO2_emissions, avg_temperature) scaler = StandardScaler() numerical_cols = ['CO2_emissions', 'avg_temperature'] df_climate[numerical_cols] = scaler.fit_transform(df_climate[numerical_cols])

print("\nNumerical columns normalized:") display(df_climate.head())

```
[26] # 4. Normalize numerical columns (CO2_emissions, avg_temperature)
scaler = StandardScaler()
numerical_cols = ['CO2_emissions', 'avg_temperature']
df_climate[numerical_cols] = scaler.fit_transform(df_climate[numerical_cols])

print("\nNumerical columns normalized:")
display(df_climate.head())
```

... Numerical columns normalized:

	CO2_emissions	avg_temperature	country	year	extreme_event_encoded
0	-0.943849	0.555558	USA	1999	0
1	0.781665	0.682045	GERMANY	2003	0
2	-1.189547	-0.742576	CANADA	2012	1
3	-0.306494	0.291049	USA	2011	1
4	-0.524881	1.796834	FRANCE	2021	0

```
[27] # 5. Generate a brief summary report
```

5. Generate a brief summary report

print("\n--- Summary Report ---")

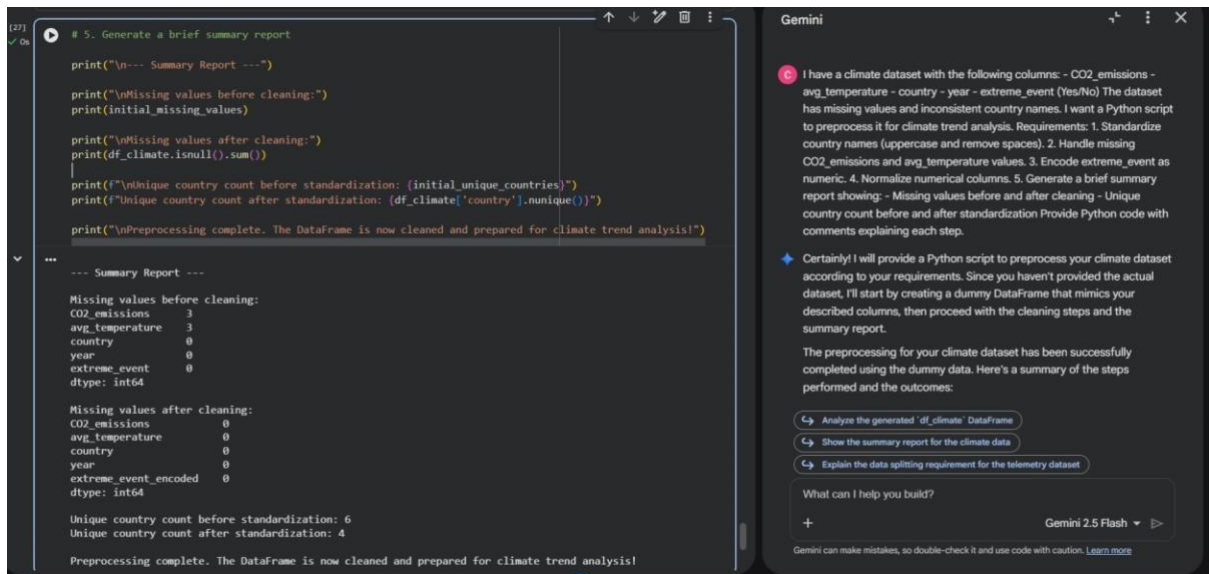
print("\nMissing values before cleaning:") print(initial_missing_values)

print("\nMissing values after cleaning:") print(df_climate.isnull().sum())

print(f"\nUnique country count before standardization: {initial_unique_countries}")

print(f"\nUnique country count after standardization: {df_climate['country'].nunique()}")

```
print("\nPreprocessing complete. The DataFrame is now cleaned and prepared for climate trend analysis!")
```



The screenshot displays a Jupyter Notebook on the left and a Gemini chat interface on the right. The Jupyter Notebook shows a Python script for preprocessing a climate dataset, including steps for handling missing values, standardizing country names, encoding extreme events, and normalizing numerical features. The output of the script is visible, showing the initial and final state of the dataset after cleaning. The Gemini chat interface on the right shows a user query about preprocessing a climate dataset and a response from the Gemini 2.5 Flash model, which provides a summary of the preprocessing steps and offers to generate a Python script for the task.

```
# 5. Generate a brief summary report

print("\n--- Summary Report ---")

print("\nMissing values before cleaning:")
print(initial_missing_values)

print("\nMissing values after cleaning:")
print(df_climate.isnull().sum())

print(f"\nUnique country count before standardization: {initial_unique_countries}")
print(f"\nUnique country count after standardization: {df_climate['country'].nunique()}")

print("\nPreprocessing complete. The DataFrame is now cleaned and prepared for climate trend analysis!")

...

--- Summary Report ---

Missing values before cleaning:
CO2_emissions    3
avg_temperature   3
country           0
year             0
extreme_event     0
dtype: int64

Missing values after cleaning:
CO2_emissions    0
avg_temperature   0
country          0
year            0
extreme_event_encoded  0
dtype: int64

Unique country count before standardization: 6
Unique country count after standardization: 4

Preprocessing complete. The DataFrame is now cleaned and prepared for climate trend analysis!
```

Explanation:

- Country names standardized to uppercase and stripped of extra spaces.
- Missing values handled with median for robustness.
- Categorical encoding for extreme events.
- Numerical features normalized.
- Data quality summary report included for Word submission.